

SUMMER INTERNSHIP PROJECT REPORT

Supervised Bit Classification for Underwater Optical Communication

Submitted by

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Abstract

Underwater Optical Wireless Communication (UWOC) offers high-speed underwater data transmission but is significantly affected by water turbidity, which causes optical signal attenuation and increases bit classification errors. Traditional fixed-threshold detection methods often fail to maintain reliable performance under varying turbidity conditions. This project investigates the use of supervised machine learning techniques for robust bit classification in UWOC systems.

A dataset containing received analog signal measurements, analog voltage values, turbidity (NTU) information, and transmitted bit labels was analyzed and preprocessed. Exploratory Data Analysis (EDA), feature engineering, scaling, and stratified train-test splitting were performed before model development. Logistic Regression, Support Vector Machine (SVM), Random Forest, and Multi-Layer Perceptron (MLP) classifiers were implemented and compared using Accuracy, Precision, Recall, F1-Score, Bit Error Rate (BER), and Inference Time. Hyperparameter tuning was carried out using GridSearchCV.

Experimental results showed that the SVM classifier achieved the best overall performance with an accuracy of **75.83%** and the lowest BER of **0.242**. Further NTU-wise analysis revealed that classification errors increased with turbidity due to greater signal distortion and overlap between bit classes. The study demonstrates the effectiveness of machine learning techniques for improving bit detection performance in UWOC systems under varying underwater conditions.

Keywords: *Underwater Optical Wireless Communication, Turbidity, Bit Classification, Support Vector Machine, Bit Error Rate, Machine Learning*

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1. Objective

The primary objective of this project is to investigate the use of supervised machine learning techniques for reliable bit classification in Underwater Optical Wireless Communication (UWOC) systems operating under varying turbidity conditions.

In underwater environments, optical signals are affected by absorption and scattering due to suspended particles present in water. As turbidity increases, the received optical signal becomes distorted, making conventional fixed-threshold detection methods less effective.

This project aims to understand the relationship between received analog signal values, turbidity levels (NTU), and transmitted binary bits. The supplied dataset contains transmitted bit labels and corresponding received signal measurements collected under different turbidity conditions. Using this dataset, a complete machine learning pipeline was developed, including data inspection, exploratory data analysis, preprocessing, feature engineering, model training, evaluation, and robustness analysis.

The project also aims to compare the performance of traditional threshold-based detection with multiple supervised machine learning algorithms such as Logistic Regression, Support Vector Machine (SVM), Random Forest, and Multi-Layer Perceptron (MLP). Performance is evaluated using Accuracy, Precision, Recall, F1-Score, Bit Error Rate (BER), and Inference Time. The final objective is to identify the most suitable classifier for robust bit recovery under changing underwater channel conditions and to analyze the limitations and failure cases of the selected model.

2. Introduction

Underwater communication plays a vital role in applications such as ocean exploration, environmental monitoring, underwater sensor networks, military surveillance, offshore resource management, and autonomous underwater vehicles (AUVs). Traditionally, underwater communication has relied on acoustic waves due to their ability to travel long distances in water. However, acoustic communication systems suffer from low data rates, high latency, limited bandwidth, and susceptibility to environmental noise. These limitations have motivated researchers to explore alternative communication technologies capable of supporting high-speed underwater data transmission.

Underwater Optical Wireless Communication (UWOC) has emerged as a promising solution for next-generation underwater communication systems. UWOC utilizes visible or near-visible light, typically in the blue-green spectrum (450–530 nm), to transmit information through water. Compared to acoustic communication, UWOC offers significantly higher bandwidth, faster transmission speeds, lower latency, and improved security. These advantages make UWOC suitable for applications requiring real-time data exchange and high-speed underwater connectivity.

Despite its advantages, UWOC systems face several challenges due to the unique characteristics of the underwater environment. One of the most important factors affecting optical communication performance is water turbidity. Turbidity refers to the concentration of suspended particles present in water and is commonly measured in Nephelometric Turbidity Units (NTU). As turbidity increases, optical signals experience greater scattering and absorption, resulting in reduced signal strength at the receiver. These effects distort the received signal and increase the probability of communication errors.

In conventional UWOC receivers, transmitted binary information is recovered using fixed-threshold detection techniques. A predefined threshold value is used to classify received signals as binary bit 0 or bit 1. While this approach is simple and computationally efficient, its performance degrades significantly when channel conditions change. Variations in turbidity cause shifts in signal amplitude distributions, making a single threshold insufficient for reliable bit classification. Consequently, the Bit Error Rate (BER) increases, reducing overall communication reliability.

Recent advances in Machine Learning (ML) have demonstrated the potential to improve communication system performance by learning patterns directly from data. Unlike threshold-based methods, supervised machine learning algorithms can automatically identify complex relationships between received signal characteristics and transmitted bit labels. By leveraging historical data, these models can adapt to varying channel conditions and provide more robust classification performance.

This project investigates the application of supervised machine learning techniques for bit classification in UWOC systems operating under different turbidity conditions. The supplied dataset contains received analog signal measurements, analog voltage values, turbidity information, and corresponding transmitted bit labels. A complete machine learning workflow was developed, including dataset inspection, exploratory data analysis, preprocessing, feature engineering, model training, hyperparameter tuning, evaluation, and robustness analysis.

Several classification algorithms, namely Logistic Regression, Support Vector Machine (SVM), Random Forest, and Multi-Layer Perceptron (MLP), were implemented and compared against traditional threshold-based detection. Model performance was evaluated using Accuracy, Precision, Recall, F1-Score, Bit Error Rate (BER), and Inference Time. Furthermore, NTU-wise analysis was conducted to study model robustness under different underwater conditions and identify failure cases.

The outcomes of this study contribute to understanding the feasibility of machine learning-based bit detection in UWOC systems and provide insights into developing more reliable underwater communication technologies for future applications.

3. Work Done

3.1 Dataset Collection and Understanding

The supplied dataset consists of received optical signal measurements collected under different turbidity conditions for a UWOC system. The objective was to classify the transmitted binary bit (0 or 1) using received signal characteristics and turbidity information.

Table 1: Dataset summary

Property	Value
Total Samples	399
Total Features	3
Target Variable	Binary Bit S
Missing Values	0
Duplicate Records	0
Invalid Labels	0

S No	binary bit	Binary bit	Analog value	Analog Voltage	NTU
1	1	1	125	0.610351563	1-2NTU
2	0	0	101	0.493164063	1-2NTU
3	1	1	123	0.600585938	1-2NTU
4	0	0	101	0.493164063	1-2NTU
5	0	0	95	0.463867188	1-2NTU
6	0	0	99	0.483398438	1-2NTU
7	1	1	142	0.693359375	1-2NTU
8	0	0	97	0.473632813	1-2NTU
9	0	0	98	0.478515625	1-2NTU
10	0	0	98	0.478515625	1-2NTU

Figure 1: Dataset overview and structure

3.2 Exploratory Data Analysis (EDA)

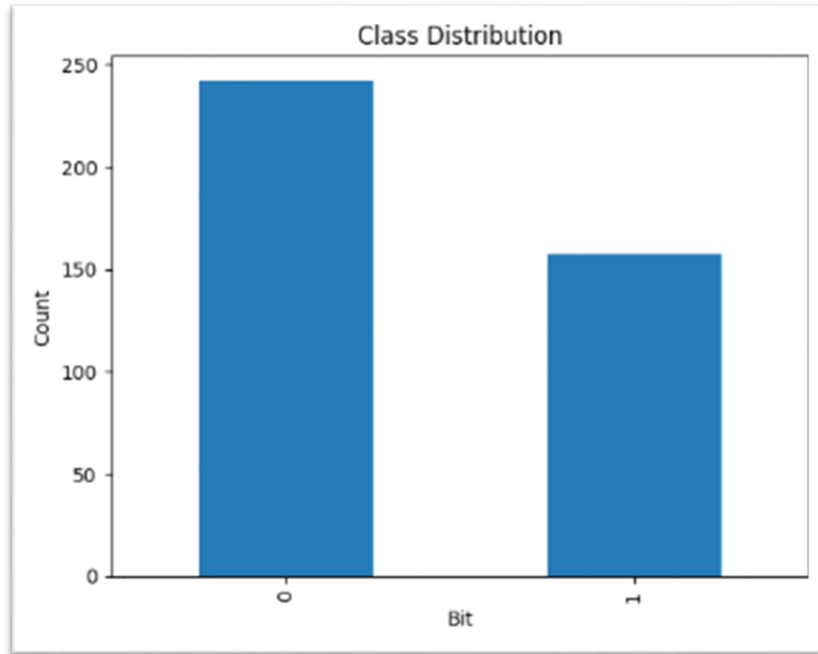
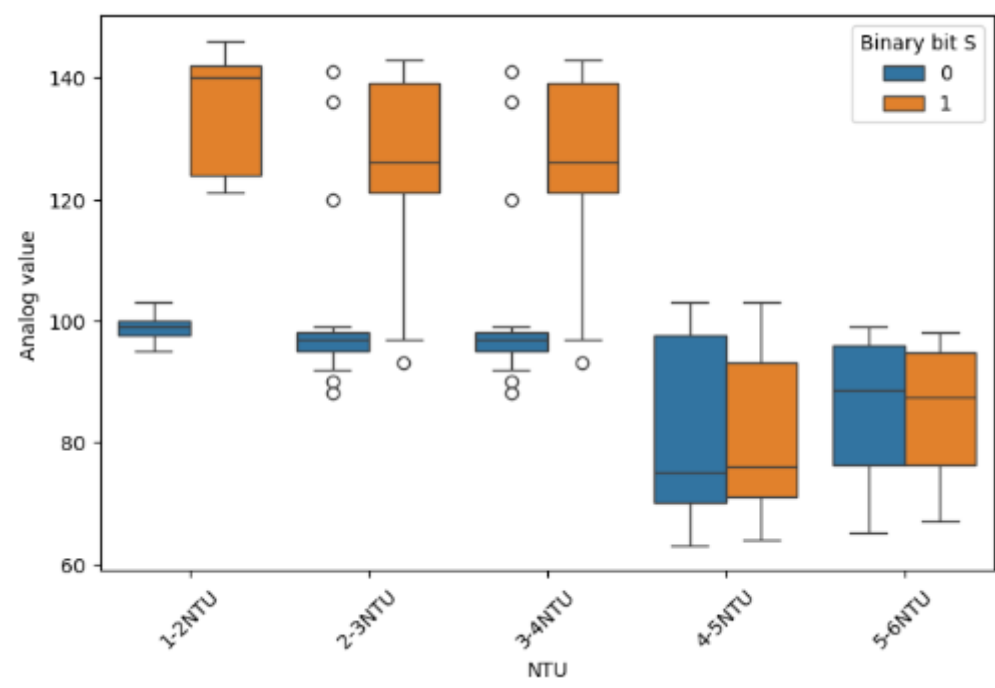
Exploratory Data Analysis was performed to understand data distribution, class balance, feature relationships, and the impact of turbidity on signal characteristics.

Key Observations

- No missing or duplicate records were found.
- The dataset contains moderate class imbalance.
- Signal distributions vary across NTU levels.
- Higher turbidity causes greater overlap between bit classes.

Table 2: Class distribution

Class	Count	Percentage
Bit 0	242	60.65%
Bit 1	157	39.35%

**Figure 2:** Class distribution plot**Figure 3:** Feature distribution plots

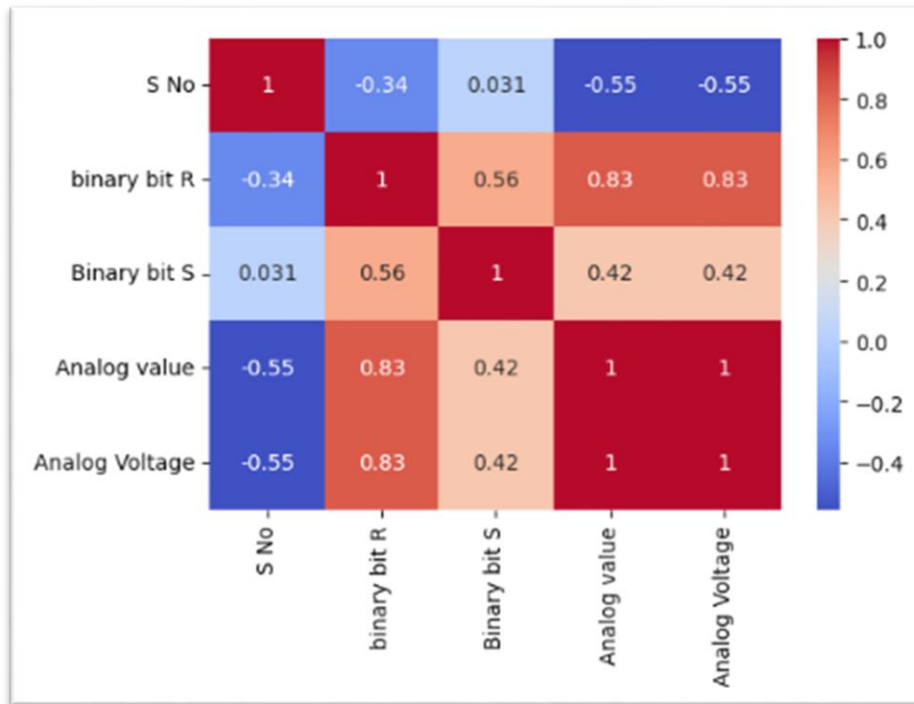


Figure 4: Correlation heatmap

3.3 Data Quality Assessment

A comprehensive quality audit was conducted to identify issues that could affect model performance.

Table 3: Data quality report

Check	Result
Missing Values	0
Duplicate Rows	0
Invalid Labels	0
Class Imbalance	Moderate
Data Leakage	Not Observed

Conclusions

- The dataset is clean and suitable for model training.
- No data cleaning was required.
- Moderate class imbalance was retained since the ratio was acceptable.

3.4 Data Preprocessing

Several preprocessing steps were applied before model training.

Preprocessing Steps

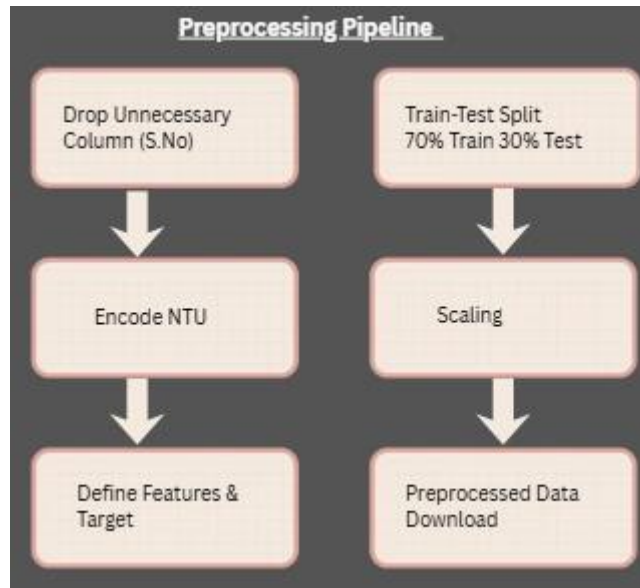
- Dataset inspection
- Target verification
- NTU encoding
- Feature scaling
- Stratified train-test split

Table 4: Train-test split

Dataset	Percentage
Training Set	70%
Testing Set	30%

Features were scaled using **StandardScaler()**, which provided the following benefits:

- Improved model convergence.
- Reduced feature dominance.
- Enhanced SVM performance.

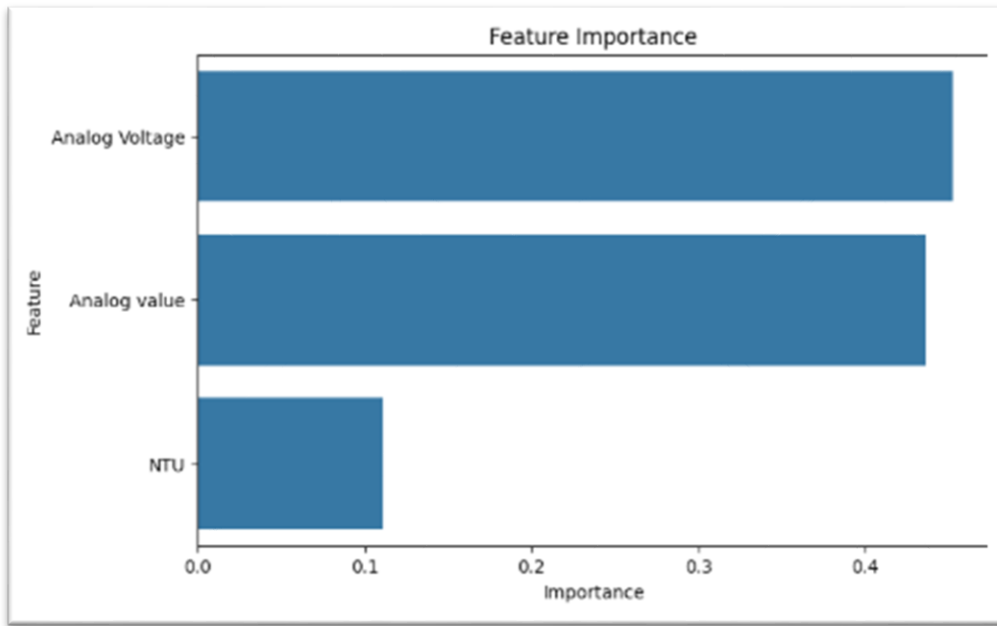
**Figure 6:** Preprocessing pipeline**3.5 Feature Engineering**

Feature engineering was performed to evaluate the contribution of each variable toward bit classification.

Table 5: Feature importance analysis

Feature	Importance
Analog Voltage	0.45
Analog Value	0.44
NTU	0.11

Analog Voltage and Analog Value contributed most significantly toward classification performance, while NTU provided useful environmental context.

**Figure 7:** Feature importance plot

3.6 Model Development

Four supervised machine learning models were implemented and compared with traditional threshold-based detection. The workflow followed was: Dataset → Preprocessing → Training → Evaluation → Comparison.

Table 6: Models implemented

Category	Model
Baseline	Fixed Threshold Detection
ML Model	Logistic Regression
ML Model	Support Vector Machine (SVM)
ML Model	Random Forest
ML Model	Multi-Layer Perceptron (MLP)

3.7 Hyperparameter Tuning

Grid Search Cross Validation (GridSearchCV) was used to identify the best-performing parameters.

Table 7: Best parameters – Logistic Regression

Parameter	Value
C	0.1
Solver	lbfgs

Table 8: Best parameters – SVM

Parameter	Value
C	0.1
Kernel	Linear

Table 9: Cross-validation settings

Parameter	Value
Method	GridSearchCV
CV Folds	5
Scoring	Accuracy

3.8 Model Evaluation

Performance was evaluated using multiple classification metrics: Accuracy, Precision, Recall, F1-Score, Bit Error Rate (BER), and Inference Time.

Table 10: Model comparison

Model	Accuracy	Precision	Recall	F1 Score	BER
Logistic Regression	74.17%	66.95%	61.70%	65.17%	0.258
SVM	75.83%	84.62%	46.81%	60.27%	0.242
Random Forest	70.83%	63.04%	61.70%	62.37%	0.292
MLP	75.00%	71.79%	59.57%	65.12%	0.250

Key Findings

- SVM achieved the highest accuracy.
- SVM produced the lowest BER.
- Logistic Regression achieved the lowest inference time.

- Random Forest showed comparatively lower performance.

3.9 NTU Robustness Analysis

Model performance was analyzed across different turbidity levels.

Table 11: NTU-wise error distribution

NTU Level	Misclassified Samples
2–3 NTU	2
3–4 NTU	2
4–5 NTU	9
5–6 NTU	16

Classification performance decreased as turbidity increased. High NTU levels caused greater signal attenuation and overlap between bit classes.

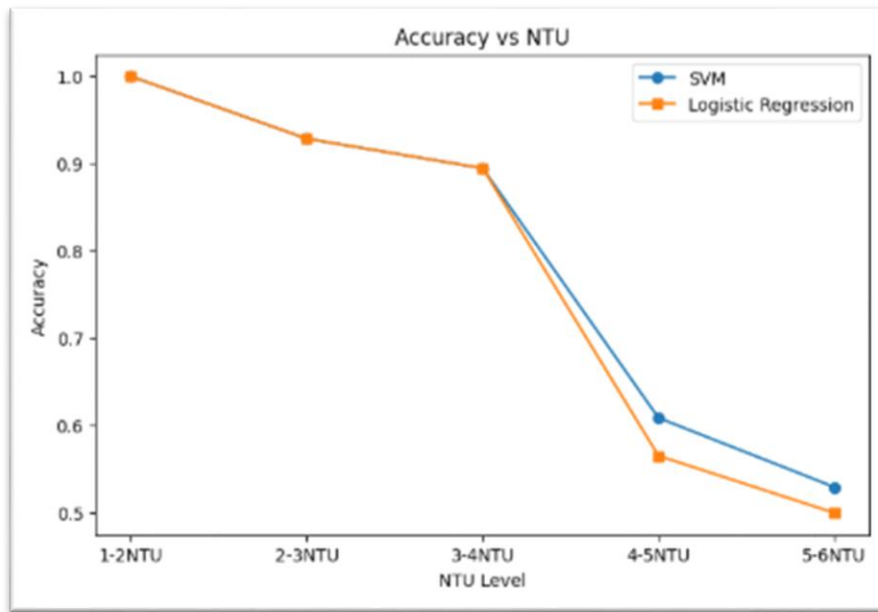


Figure 12: NTU-wise accuracy analysis

3.10 Error Analysis

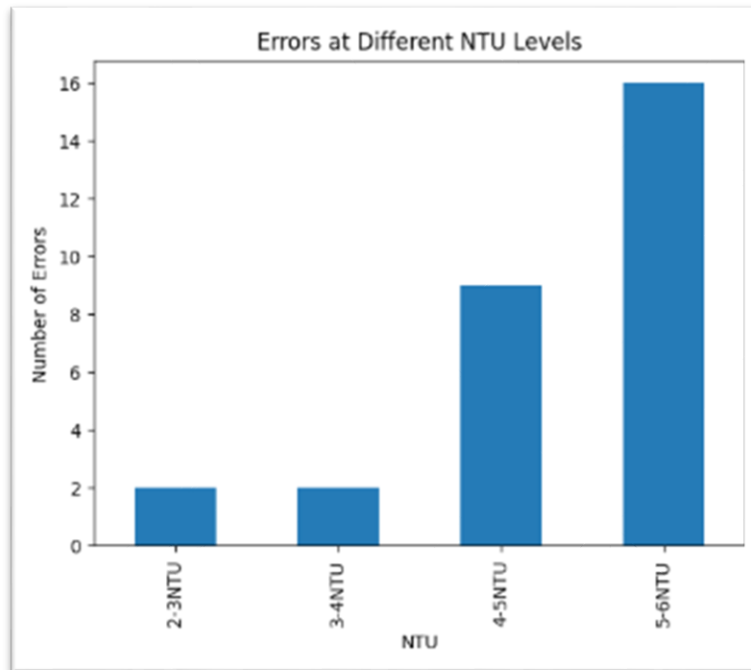
The final SVM model was analyzed to identify failure cases.

Table 12: Prediction summary (SVM)

Metric	Value
Correct Predictions	91
Incorrect Predictions	29
Accuracy	75.83%

Findings

- Most errors occurred under high turbidity conditions.
- Signal overlap was the primary cause of misclassification.
- Low NTU conditions showed significantly better classification performance.

**Figure 14:** Misclassification analysis**3.11 Final Model Selection**

Based on overall performance, the Support Vector Machine (SVM) was selected as the final model, for the following reasons:

- Highest accuracy (75.83%)
- Highest precision (84.62%)
- Lowest BER (0.242)
- Better robustness across varying NTU conditions

4. Future Work

Although the proposed machine learning framework demonstrated promising results for bit classification in UWOC systems, several opportunities exist for further improvement and extension of the work.

Larger and more diverse dataset. The current study was conducted on a dataset containing 399 samples collected under a limited range of turbidity conditions. Increasing the dataset size and incorporating additional environmental scenarios would improve the generalization capability and robustness of the models.

Advanced and ensemble learning. Future research can investigate techniques such as XGBoost, LightGBM, CatBoost, and stacking-based approaches. These models may capture complex relationships between signal characteristics and transmitted bits more effectively, potentially leading to improved accuracy and lower BER.

Deep learning architectures. Models such as Artificial Neural Networks (ANNs), Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs) could be evaluated for their ability to learn nonlinear signal patterns and adapt to changing underwater channel conditions, particularly when larger datasets become available.

Additional input parameters. The current work primarily utilized received signal measurements and turbidity information. Future studies may incorporate communication distance, water temperature, salinity, optical wavelength, and transmitter power, which influence underwater optical signal propagation.

Real-time deployment. The models developed in this study were evaluated using offline experimental data. Future work can focus on implementing the trained models on embedded systems or edge devices to evaluate computational efficiency, latency, and practical feasibility in operational environments.

Adaptive and online learning. Such techniques would allow the classifier to continuously update itself as underwater channel conditions change, which is particularly useful in dynamic aquatic environments where turbidity varies over time.

Hybrid approaches. Combining traditional signal processing methods with machine learning-based classification may leverage the advantages of both to achieve higher reliability and robustness under challenging conditions.

Overall, the findings of this project provide a strong foundation for future research on intelligent and adaptive UWOC systems, with the potential to improve the reliability and efficiency of underwater optical communication in real-world applications.

5. Conclusion

This project investigated the application of supervised machine learning techniques for bit classification in UWOC systems operating under varying turbidity conditions. The study was motivated by the limitations of conventional fixed-threshold detection methods, which become less reliable as water turbidity increases and causes significant distortion in received optical signals.

A complete machine learning workflow was developed, including dataset inspection, exploratory data analysis, data preprocessing, feature engineering, model development, hyperparameter tuning, evaluation, and error analysis. Data quality assessment confirmed that the dataset contained no missing values, duplicate records, or invalid labels, making it suitable for model development.

Multiple classification models, namely Logistic Regression, SVM, Random Forest, and MLP, were implemented and compared using Accuracy, Precision, Recall, F1-Score, BER, and Inference Time. Among all evaluated models, the Support Vector Machine achieved the best overall performance with an accuracy of **75.83%** and the lowest BER of **0.242**, while Logistic Regression demonstrated the lowest inference time.

NTU-wise robustness analysis showed that classification performance decreases as turbidity increases due to stronger optical signal attenuation and greater overlap between bit classes. Error analysis further revealed that the majority of misclassifications occurred under higher turbidity conditions, highlighting the challenges of reliable underwater optical communication in highly turbid environments.

Overall, the results demonstrate that supervised machine learning methods can improve bit classification performance compared to traditional threshold-based approaches and provide a more robust solution for UWOC systems operating under varying environmental conditions. The findings establish a foundation for future research on intelligent and adaptive underwater communication systems.

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This internship has been a valuable learning experience that enhanced my understanding of machine learning, data analysis, and research methodology, and I am grateful to everyone who contributed to its successful completion.

References

- [1]Z. Zeng, S. Fu, H. Zhang, Y. Dong, and J. Cheng, “A Survey of Underwater Optical Wireless Communications,” *IEEE Communications Surveys & Tutorials*, vol. 19, no. 1, pp. 204–238, 2017.
- [2]H. Kaushal and G. Kaddoum, “Underwater Optical Wireless Communication,” *IEEE Access*, vol. 4, pp. 1518–1547, 2016.
- [3]C. M. Bishop, *Pattern Recognition and Machine Learning*. New York, NY, USA: Springer, 2006.
- [4]T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2nd ed. New York, NY, USA: Springer, 2009.
- [5]F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [6]C. Cortes and V. Vapnik, “Support-Vector Networks,” *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.